3. Design a CPU Scheduling Program with C Using First Come First Served (FCFS)

Code - Part 1 (Lines 1–34)

```c
1  #include <stdio.h>
2
3  int main()
4  {
5      int A[100][4];
6      int i, j, n, total = 0, index, temp;
7      float avg_wt, avg_tat;
8
9      printf("Enter number of process: ");
10     scanf("%d", &n);
11
12     printf("Enter Burst Time:\n");
13     for (i = 0; i < n; i++)
14     {
15         printf("P%d: ", i + 1);
16         scanf("%d", &A[i][1]);
17         A[i][0] = i + 1; // Process ID
18     }
19
20     // Sorting based on Burst Time (SJF)
21     for (i = 0; i < n; i++)
22     {
23         index = i;
24         for (j = i + 1; j < n; j++)
25         {
26             if (A[j][1] < A[index][1])
27                 index = j;
28         }
29
30         // Swap Burst Time
31         temp = A[i][1];
32         A[i][1] = A[index][1];
33         A[index][1] = temp;
34
```

Code - Part 2 (Lines 35–64)

```
35          // Swap Process ID
36          temp = A[i][0];
37          A[i][0] = A[index][0];
38          A[index][0] = temp;
39      }
40
41      // Waiting Time
42      A[0][2] = 0;
43      for (i = 1; i < n; i++)
44      {
45          A[i][2] = 0;
46          for (j = 0; j < i; j++)
47          {
48              A[i][2] += A[j][1];
49          }
50          total += A[i][2];
51      }
52
53      avg_wt = (float)total / n;
54
55      // Turnaround Time
56      total = 0;
57
58      printf("\nP\tBT\tWT\tTAT\n");
59      for (i = 0; i < n; i++)
60      {
61          A[i][3] = A[i][1] + A[i][2];
62          total += A[i][3];
63
64          printf("P%d\t%d\t%d\t%d\n",
```

Code - Part 3 (Lines 65–77)

```
65                  A[i][0],
66                  A[i][1],
67                  A[i][2],
68                  A[i][3]);
69      }
70
71      avg_tat = (float)total / n;
72
73      printf("\nAverage Waiting Time = %.2f", avg_wt);
74      printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
75
76      return 0;
77  }
```

Output

```
Enter number of process: 3
Enter Burst Time:
P1: 2
P2: 3
P3: 4

P        BT       WT       TAT
P1       2        0        2
P2       3        2        5
P3       4        5        9

Average Waiting Time = 2.33
Average Turnaround Time = 5.33


...Program finished with exit code 0
Press ENTER to exit console.
```